

2. Requirement Analysis Phase

2.1 Introduction

The Requirement Analysis Phase identifies, analyzes, and documents the requirements of ComicCraft: AI Comic Story Creator using Gemini Models. It focuses on understanding the problems faced by users in turning story ideas into comics and defining the features needed to solve them. ComicCraft is designed as an intelligent creative system that uses Google Gemini models to write comic scripts, generate panel artwork, and assemble finished comics.

2.2 Problem Analysis

Comic creation is difficult for beginners. Users may have a story but lack drawing skills, may not know how to break a story into panels, may find that characters look different in every generated image, and may struggle with lettering and layout. The major problems identified are:

- Lack of drawing and design skills among story writers.
- Difficulty in dividing a story into well-paced panels.
- Inconsistent character appearance across AI-generated images.
- Time-consuming manual lettering, bubble placement, and page layout.
- Difficulty in matching a single art style across all panels.
- No easy way to edit or regenerate individual panels.
- Difficulty in saving, sharing, and exporting finished comics.

2.3 Objectives of Requirement Analysis

- To identify the needs and expectations of users.
- To define the major functionalities of ComicCraft.
- To identify the required hardware and software resources.
- To determine functional and non-functional requirements.
- To define requirements for Gemini-based text and image integration.
- To provide a clear foundation for design and development.

2.4 Functional Requirements

2.4.1 User Registration and Authentication

- User registration and secure login.
- User profile management (name, email, default art style).
- Logout functionality.
- Basic authentication and input validation.

2.4.2 Story Input

- Accept a story idea, genre, tone, target audience, art style, and number of panels.
- Optional character names and descriptions supplied by the user.
- Option to start from a template or sample prompt.

2.4.3 AI Story and Script Generation

- Generate a title, short synopsis, character list, and panel-by-panel script using Gemini.
- Each panel contains a scene description, dialogue, caption, and camera angle.
- Allow the user to edit or regenerate the script before drawing.

2.4.4 Character Management

- Create and store character profiles (appearance, clothing, personality).
- Include the character description in every relevant panel prompt to maintain consistency.
- Optional AI-generated character reference sheet.

2.4.5 Panel Image Generation

- Generate artwork for each panel using a Gemini image generation model.
- Apply the selected art style consistently.
- Regenerate any single panel with a modified prompt.

2.4.6 Speech Bubbles and Captions

- Overlay dialogue and captions on panels.
- Edit text, move, and resize bubbles.
- Choose bubble style and font.

2.4.7 Layout and Export

- Arrange panels into comic pages with selectable layouts.
- Preview the full comic.
- Export as PDF or PNG images.

2.4.8 Gallery and History

- Store saved comics, scripts, and character sheets.
- Open, edit, duplicate, or delete previous comics.

2.5 Non-Functional Requirements

Requirement	Description
Usability	Simple, clean, responsive interface usable without drawing or technical skills.
Performance	Script generation and image generation should complete within a reasonable time, with progress indicators.
Security	Secure authentication, protected APIs, and the Gemini API key kept on the server.
Privacy	Users can access only their own comics and characters.
Content Safety	Prompts and outputs are filtered for unsafe or inappropriate content using Gemini safety settings.
Reliability	Accurate storage of data and graceful handling of API errors, timeouts, and blocked generations.
Scalability	Architecture should allow new art styles, features, and users to be added.
Maintainability	Modular, well-organized code that is easy to test and extend.

2.6 Hardware Requirements

- Computer or laptop with minimum 4 GB RAM and 10 GB free storage.
- Internet connection (required for Gemini API access).
- Keyboard and mouse; smartphone or computer for accessing the application.

2.7 Software Requirements

Component	Technology
Frontend	HTML, CSS, JavaScript (Canvas / SVG for speech bubbles)
Backend	Python, FastAPI
Database	SQLite / PostgreSQL
Image Storage	Local file storage or cloud object storage
AI / LLM	Google Gemini API (text model for scripts, image model for panel art)
Authentication	JWT
Export	Pillow and ReportLab for image composition and PDF export
IDE	Visual Studio Code
Version Control	Git and GitHub
Operating System	Windows / Linux / macOS

2.8 Expected Outcome

The phase produces a well-defined set of requirements that serves as the foundation for the planning, design, development, testing, and demonstration phases. ComicCraft is expected to give users a convenient platform for turning story ideas into finished, illustrated comics with AI support.